

# Kunal Maheshwari

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## SUMMARY

OPT-EAD holder with full-time work authorization and over 4 years of experience as a Software Developer and Data Engineer, specializing in building scalable microservices and distributed data pipelines. Possesses expertise in Java, Python, Spark, SQL, AWS, and Azure, with a strong foundation in CI/CD automation, architecture design, and large-scale data platforms. Proven ability in backend development and optimization techniques, driving efficient software engineering solutions.

## EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025  
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020  
Bachelor of Technology | Computer Science and Engineering

## EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, managing over 50,000 daily API requests with 99.99 percent SLA uptime.
- Automated CI/CD pipelines with Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x and enhancing monitoring, reducing issue resolution time by 20 percent.
- Spearheaded a major EDX platform upgrade, optimizing application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing, demonstrating proficiency in software engineering principles.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Developed Java and EJB-based enterprise applications and RESTful web services for automotive clients, contributing to robust backend development.
- Achieved 90 percent system performance improvement through SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, leveraging optimization techniques.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices, fostering a collaborative software engineering environment.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage, a key aspect of software engineering.
- Supported release planning and sprint execution using JIRA within Agile development teams, demonstrating logistics and workflow management.

## PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time, showcasing expertise in backend development and large-scale data platforms.
- Built an interactive Flask web application for data exploration with dynamic filtering and visualization, utilizing Python and optimization techniques.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, demonstrating architecture design capabilities.

Privacy-Preserving AI Email Client (Capstone Project - T5, LoRA, Python) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, integrating C++ principles for efficiency.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, touching upon sensor data processing concepts for input handling.

## SKILLS

- Programming Languages: Proficient in Java, Python, SQL, C++, and Go, with a strong understanding of data structures.
- Frameworks and Libraries: Extensive experience with Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, and JUnit, enhancing backend development capabilities.
- Web Services: Expertise in designing and developing REST API and SOAP API solutions.

- Cloud Platforms: Skilled in AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), and Google Cloud Platform, alongside infrastructure automation using Terraform.
- Databases: Proficient with PostgreSQL, Oracle Database, IBM DB2, and RDS, supporting data management and optimization techniques.
- Tools and Technologies: Adept with Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and leveraging logistics tools for workflow management.
- Concepts: Strong grasp of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/ CD, Microservices, Large- scale data platforms, Architecture Design, Sensor Data Processing, and backend development.
- Data Analysis and Visualization: Experienced with Tableau and QuickSight for insightful data representation.